Unit 5
Applications and Impacts of Computing

Machine learning · neural nets · deep learning · threats · equity

LAYERS & WEIGHTS | PHISHING & 2FA | ENCRYPTION | EQUAL ACCESS VS EQUITY

A phone that unlocks a face is impressive. A class that can name the threat model is safer.

1. Three nested ideas

Table 1. Do not treat these as synonyms

Term	Role
Machine learning	The field: a model studies data and then predicts or decides
Neural network	One kind of model — layers of nodes inspired by (not copying) a brain
Deep learning	A method: train a neural net that has *many* hidden layers

Face unlock, speech assistants, photo search, shopping recommendations, medical imaging, translation — all sit on this stack. Pakistan's National AI Policy conversation (2025 in the textbook) is the local "why we bother".

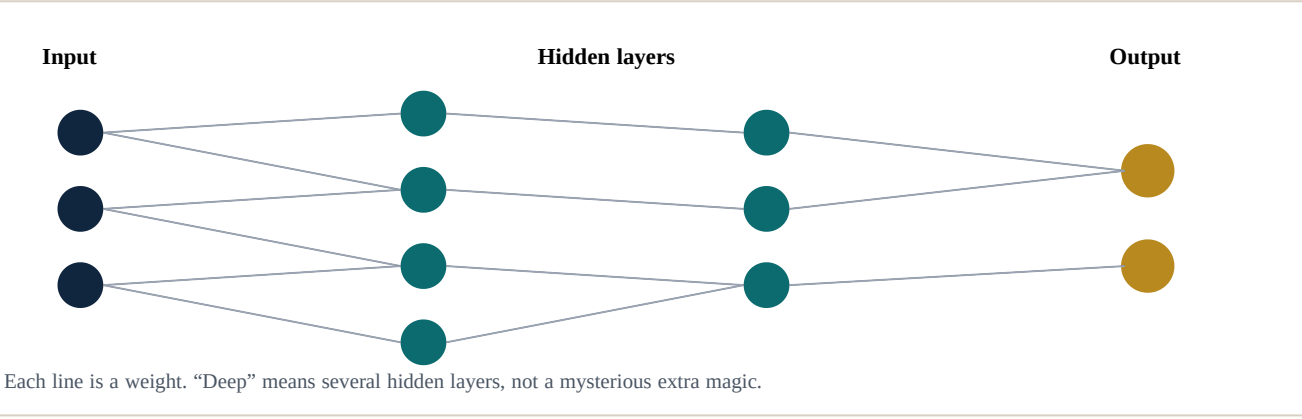


Figure 1. A tiny feed-forward net. Real image models have millions of weights; the picture stays the same.

1.1 Components of a neural net

- **Input layer** — raw pixels, samples of sound, word IDs, sensor numbers.
- **Hidden layer(s)** — where patterns are built (edges → eyes → face).
- **Output layer** — the prediction ("this is Ayesha" / "spam").
- **Weights** and **bias** — knobs the training loop turns.
- **Activation** — a non-linear gate so the net can learn curves, not only straight lines.
- **Loss** — a number that says how wrong this guess was; training tries to shrink it.

$$z = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b, \qquad \hat{y} = f(z), \qquad \text{train by reducing } \text{Loss}(y, \hat{y})$$

1.2 What "deep" adds

Large data, many hidden layers, automatic feature learning (you do not hand-draw "eye detectors"), repeated training passes, a learning algorithm, and serious compute (often a GPU). Applications: face and image recognition, speech, NLP (translators, chatbots), medical scans, recommendations, driving assistance, fraud alerts, captions and speech-to-text as assistive tech.

2. Secure collaboration and data protection

Group work on an LMS is only "collaboration" if the wrong person cannot join and the right person's file cannot vanish.

AUTHENTICATION
Password, OTP, biometric — prove you are you.

ACCESS CONTROL
View / edit / share rights. Not everyone is an owner.

ACCOUNTABILITY
Logs: who changed the doc at 22:14?

PRIVATE CHANNELS
Official LMS / school email beat a public random link.

Encryption turns plaintext into ciphertext.

$$\text{plaintext} \xrightarrow[\text{key}]{\text{encrypt}} \text{ciphertext} \xrightarrow[\text{key}]{\text{decrypt}} \text{plaintext}$$

- **Symmetric:** same secret both ways — fast; key-sharing is the hard part.
- **Asymmetric:** public key to lock, private key to unlock — how HTTPS starts.

Also: strong passwords, backup/recovery (the Sukkur crash story in the book), firewall, antivirus.

3. Threats and the boring defences that work

Table 2. Name the threat, then name the shield

Threat	What it looks like	Mitigation
Malware	Virus, worm, ransomware	Antivirus, do not sideload junk
Phishing	“Update your fee portal” email (Hyderabad college example)	Check the real domain; never send a password
Unauthorised access	Shared passwords, open lab accounts	Unique passwords, **2FA**
Data / identity theft	Stolen marks file, CNIC photos	Encryption, least privilege
DoS	Service flooded until it dies	Provider protections, rate limits
Human error	Wrong share link, no backup	Training, official tools, rehearsed restore

4. Equity is not the same as equal access

Equal access hands everyone the same Zoom link. **Equity** adds what a particular learner needs so that link is actually usable: captions, a recorded lecture for the student on a 2G connection, a mobile-friendly page, a screen reader. Rural Sindh hearing-support in the textbook is an equity story, not a gadget story.

Collaboration tools: Google Docs, Zoom, Microsoft Teams, Google Classroom / LMS, email, and WhatsApp used carefully (no leaked marksheets in a 200-person group).

Self-check

1. Is deep learning a different field from machine learning, or a method inside it?
2. Public key encrypts a message to you. Which key decrypts it?
3. A teacher posts the same PDF to every student. Equal access or equity? What would equity add for a student with low vision?

Answers. (1) A method inside ML (training deep nets). (2) Your private key. (3) Equal access; equity would add a tagged/large-type or audio version and a high-contrast theme.

Recap

1. ML learns; a neural net is a layered model; deep learning trains many hidden layers.
2. Input / hidden / output + weights, bias, activation, loss.
3. Protect collaboration with authentication, access control, logs, encryption, backups.
4. Threats: malware, phishing, theft, DoS, human error. Shields: 2FA, firewall, awareness.
5. Equal access $\neq$ equity. Design the extra support on purpose.

Redraw Figure 1 and label a phishing email with three red flags.